Activity 4.3: Engagement Scoring Logic

1. Activity Overview & Objective

This activity focuses on transforming fused multimodal signals into interpretable **engagement scores**. Rule-based logic and predefined thresholds are applied to assess engagement levels across different time segments of the session. Factors such as sustained visual presence, movement frequency, inactivity duration, and speech activity intensity are collectively considered.

> **Temporal Engagement Profiling:** Engagement scores are computed at regular intervals to generate a temporal engagement profile rather than a single aggregated value. This approach allows identification of attention fluctuations, peak engagement periods, and extended disengagement phases. The scoring logic is designed to be transparent and adjustable, enabling fine-tuning based on classroom requirements.

2. Multimodal Scoring Factors & Threshold Matrix

To avoid black-box ambiguity, the scoring engine evaluates four primary behavioral dimensions extracted from the synchronized multimodal stream at fixed time segments. Each factor is evaluated against configurable classroom thresholds before computing the interval composite score:

Scoring Factor	Source Modality	Predefined Threshold Logic	Impact on Interval Engagement Score
Sustained Visual Presence	Vision Module	Continuous face/body detection ratio ≥ 0.80 within the active time window.	Establishes baseline attendance stability and positive attention weight.
Movement Frequency	Vision Module	Moderate pose transitions (0.15–0.55) vs. excessive restlessness (> 0.65).	Rewards natural attentive posture shifts; penalizes erratic fidgeting or distraction.
Inactivity Duration	Vision + Audio	Extended static posture combined with zero acoustic interaction (> 45s).	Applies progressive decay penalty to flag passive zoning-out or prolonged disengagement.
Speech Activity Intensity	Audio Module	Structured vocal energy and VAD ratio aligned with classroom dialogue thresholds.	Boosts interval scores during active Q&A, lectures, or collaborative discussions.

3. Reference Implementation (`backend/core/engagement_scorer.py`)

The `EngagementScorer` class computes interval-level engagement scores and builds a full temporal session profile—exposing peak engagement windows, attention fluctuations, and extended disengagement phases for downstream teaching suggestions, student analytics, and session summaries.

```
# File: backend/core/engagement_scorer.py

class EngagementScorer:
    def __init__(self, config: dict = None):
        # Transparent, adjustable weights & thresholds for classroom fine-tuning
        self.weights = (config or {}).get("weights", {
            "presence": 0.35,
            "movement": 0.20,
            "inactivity_penalty": 0.20,
            "speech_intensity": 0.25,
        })
        self.peak_threshold = 75.0
        self.disengaged_threshold = 40.0

    def compute_interval_score(self, segment: dict) -> dict:
        """Computes an interpretable engagement score (0-100) for a single time
segment."""
        presence = segment.get("sustained_presence", 0.0)        # Normalized 0.0
- 1.0
        movement = segment.get("movement_frequency", 0.0)        # Normalized 0.0
- 1.0
        inactivity = segment.get("inactivity_duration_ratio", 0.0)
        speech = segment.get("speech_activity_intensity", 0.0)

        movement_score = 1.0 - abs(movement - 0.35)
        raw_score = (
            (presence * self.weights["presence"]) +
            (max(0.0, movement_score) * self.weights["movement"]) +
            (speech * self.weights["speech_intensity"]) -
            (inactivity * self.weights["inactivity_penalty"])
        ) * 100.0

        score = round(max(0.0, min(100.0, raw_score)), 2)
        state = (
            "Peak Engagement" if score >= self.peak_threshold
            else "Disengaged" if score <= self.disengaged_threshold
            else "Moderate Engagement"
        )
        return {"timestamp": segment.get("timestamp"), "score": score, "state":
state}

    def generate_temporal_profile(self, segments: list) -> dict:
        """Generates a temporal engagement profile across regular session
intervals."""
        timeline = [self.compute_interval_score(seg) for seg in segments]
        scores = [item["score"] for item in timeline] or [0.0]

        return {
            "temporal_profile": timeline,
            "mean_score": round(sum(scores) / len(scores), 2),
            "peak_periods": [t["timestamp"] for t in timeline if t["state"] ==
"Peak Engagement"],
            "disengagement_phases": [t["timestamp"] for t in timeline if
```

4. Downstream Integration

- **Teaching Suggestions:** Real-time detection of attention fluctuations or extended disengagement phases triggers immediate pedagogical prompts for the instructor.
- **Student Analytics & Session Summaries:** Interval scores and peak engagement periods are persisted to generate post-session trend visualizations and comprehensive participation reports.